01

System overview and business outcomes

The system converts maintenance messages from existing resident channels into prepared, visible work. It identifies what requires attention, obtains missing operational details, and presents the next decision while authorized employees retain control over dispatch, spending, sensitive cases, appointments, and closure.

How the operating process changes

CURRENT PROCESS

1. Requests arrive across separate channels →
2. Staff read and interpret every message →
3. Staff chase missing details →
4. Staff search records, policies, and vendors →
5. Staff assemble the decision manually

SYSTEM-ASSISTED PROCESS

1. Existing channels feed one controlled intake →
2. Each message is linked, structured, and checked →
3. The resident receives specific approved questions →
4. Context and eligible options are prepared →
5. The manager receives one decision package

Operational outcomes to measure

OUTCOME	HOW IT IS MEASURED	WHY IT MATTERS OPERATIONALLY
Faster informed decisions	Report-to-manager-ready time	Measures preparation time without confusing it with repair time.
Better emergency visibility	Acknowledgment time; missed and unnecessary escalations reviewed separately	Shows whether dangerous conditions reach the right person without making every message an alert.
Less coordination rework	Repeat-contact rate; clarification rate and usefulness	Shows whether residents and staff need fewer repeated conversations to establish the basic facts.
Fewer duplicate actions	Duplicate work orders and vendor contacts	Tests whether replies, repeated messages, and repeated approvals remain attached to one request.
Clearer ownership	Age of unowned requests and exceptions	Shows whether waiting work always has a visible owner and next action.
Lower intake workload	Staff minutes spent reading, copying, looking up, and preparing	Measures capacity returned to coordination, vendor performance, and resident service.

Operating boundary The system prepares work and records each handoff. Authorized people decide vendor dispatch, vendor selection, spending approval, sensitive or legal matters, appointment commitments, reimbursement, and final closure.

02

How one request moves through the system

The same request remains one auditable record from the resident's first message through the final human-verified closure.

RUNNING EXAMPLE *"There is water under the kitchen sink again and it is spreading toward the dishwasher. I turned the tap off but it is still coming out. My dog is in the unit and I can let someone in."*

1

Receive and connect the request

Preserve the resident's exact words. Verify or resolve the resident, property, unit, conversation, and related open work. Detect repeated messages and follow-ups before creating anything new.

VISIBLE RESULT
One intake record linked to the correct context

2

Interpret the issue and apply policy

AI INTERPRETATION
Structures the symptom, likely trade, urgency recommendation, missing details, access permission, pet information, and sensitive-case indicators.

FIXED POLICY RULES
Apply safety, legal, notification, authority, and business rules separately. Rules may raise urgency but cannot lower it.

EXAMPLE RESULT
Spreading water triggers the company's water-escalation policy

3

Clarify and prepare

Ask only for missing operational details. Prepare the [Client PMS] record, internal summary, vendor work description, and eligible vendor options. Do not interrupt the manager while safe clarification remains possible.

EXAMPLE RESULT
Access and pet details are already complete; plumbing options are prepared

4

Present the human decision

Show the original message, prepared summary, escalation reason, property context, authority requirements, and permitted actions. The authorized employee may hold, edit, reclassify, assign manually, request owner approval, or select a vendor.

HUMAN CHECKPOINT
The manager selects the approved after-hours plumber

5

Execute and track the confirmed action

Claim the action once, write the confirmed result to [Client PMS], and contact the selected vendor through [Vendor Channel]. Update the resident only after delivery succeeds. Track response, appointment state, progress, verification, and human closure.

EXAMPLE RESULT
The vendor receives one request; the resident receives a factual update with no invented ETA

Only three exception branches are needed at this level

Missing information
Resident clarification
↲ Reassess the same request

Sensitive, unreadable, or uncertain
Assigned human review
No automated substantive action

No eligible vendor or failed contact
Visible coordinator exception
Request remains open

03

Integration into the client's existing business

The system operates behind the channels and systems already used by residents, staff, owners, and vendors. The final design depends on the capabilities and permissions of the prospect's actual software.

Four-layer operating model

1 **Existing resident channels** How requests already enter the business	[Resident Portal] [Maintenance Email] [SMS Provider] [Phone / Voicemail System]
Authorized events and replies move in both directions	
2 **Maintenance coordination layer** Prepares work and controls handoffs	Capture and context; AI interpretation + fixed policy; Conversation and request state; Retry, audit, and exceptions; Human approval routing
Confirmed facts and prepared actions are synchronized	
3 **Existing business systems** Where authoritative work and decisions remain	[Client PMS] [Manager Channel] [Vendor Channel] [Owner Approval Process] [Reporting Tool]
People continue using familiar systems	
4 **Existing users** No new process is assumed until the client's tools are reviewed	Residents; Coordinators; Managers; Owners; Vendors; Operations leadership

What each group experiences

GROUP	WHAT REMAINS FAMILIAR	WHAT THE SYSTEM ADDS
Residents	Current portal, email address, text number, or phone line	Receipt, focused clarification, and factual status from confirmed records
Coordinators	[Client PMS] work list and communication history	Prepared fields, summaries, priorities, missing details, ownership, and visible exceptions
Managers	[Manager Channel] and linked business record	One decision package with permitted actions and the reason each action is available
Vendors	[Vendor Channel], portal, PMS assignment, or email	Complete approved work descriptions and tracked delivery state
Owners	[Owner Approval Process]	Prepared scope and authority context without changing the management agreement
Executives	[Reporting Tool]	Normalized volume, response, aging, exception, correction, and vendor-outcome events

Connection pattern depends on the current stack

STARTING POINT	PRODUCTION CONNECTION	EMPLOYEE EXPERIENCE
Mature PMS APIs and webhooks	Controlled two-way work-order, note, task, assignment, and status synchronization	Prepared work appears and updates inside [Client PMS]
Limited PMS write access	Roster synchronization plus prepared tasks, controlled imports, or staff-approved creation	Staff keep the PMS as the business record; some creation remains manual
Mailbox or portal notifications	Secure message ingestion with outcomes returned through the PMS, task process, or shared inbox	Residents and staff keep their existing communication process

Integration rule [Client PMS] remains authoritative for residents, properties, leases, work orders, assignments, and final status. The coordination layer keeps processing history, retries, duplicate protection, model output, and audit events, then reconciles confirmed business changes back to the authoritative system.

04

Operating controls and client configuration

Production readiness depends on explicit authority, visible failure behavior, and configuration based on the client's actual policies and systems.

Automatic preparation versus human authority

The system may prepare automatically

- Receive and preserve authorized maintenance reports
- Link residents, units, conversations, and open work
- Detect duplicates and attach follow-ups
- Recommend urgency, trade, and missing information
- Apply fixed escalation and business rules
- Ask approved clarification questions
- Prepare work-order notes and communication drafts
- Propose eligible vendors without contacting them
- Retry temporary technical failures within policy

Authorized people retain control

- Whether a vendor is contacted
- Which vendor is selected
- Spend above the applicable authority limit
- Accommodation, discrimination, legal, insurance, or liability matters
- Changing an emergency classification downward
- Appointments, completion times, reimbursements, or rent credits
- Safety rules, vendor eligibility, and communication policy
- Verification and final work-order closure

Failure behavior remains visible to the business

SITUATION	SYSTEM RESPONSE	WHAT REMAINS VISIBLE
AI unavailable or unusable	Preserve the request, run fixed rules, and assign human review	Original message, failed step, rules result, and owner
PMS or communication channel unavailable	Queue the action, retry within policy, and expose the unsynchronized state	Pending action, attempts, failure reason, and next owner
Vendor contact fails	Keep the request open, notify the manager, and make no resident claim	Selected vendor, delivery attempts, error, and next action
Duplicate or stale approval	Accept only the first valid claim and reject the repeated action	Successful actor plus every rejected attempt
Resident identity is uncertain	Reveal no property or work-order information and route to verification	Unverified conversation and assigned task
Data differs between systems	Create a reconciliation task instead of overwriting silently	Both values, source systems, timestamps, and responsible owner

Client-specific configuration inputs

1 **Intake and identity**
Current resident channels, verified identities, property and unit lookup, and conversation history.

2 **PMS and records**
Connection capabilities, permitted reads and writes, work-order stages, and authoritative fields.

3 **Priority and safety**
Emergency definitions, habitability policy, response targets, sensitive-case handling, and escalation contacts.

4 **Notification policy**
On-call roles, quiet hours, immediate alerts, silent tasks, reminders, and daily reporting.

5 **Vendors**
Trade, property coverage, hours, priority, compliance, contact routes, and manual-assignment rules.

6 **Authority**
Manager limits, owner approval conditions, emergency exceptions, and permitted actions by role.

7 **Communication**
Approved resident, manager, owner, and vendor channels, templates, delivery tracking, and opt-out rules.

8 **Data and service**
Hosting, access, retention, deletion, security, audit, monitoring, recovery, support, and reporting requirements.

The final design is produced from the client's actual operating rules and software capabilities. No integration, automatic action, or authority boundary is assumed merely because it appears in this reference model.